Project Title

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Roll No.: **BS2516**

April, 2026

Abstract

Project Abstract

1 Introduction

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2 Data Description

The dataset used in this study is the Palmer Penguins dataset. It contains morphological measurements of penguins observed in the Palmer Archipelago, Antarctica. It includes data for three species: Adelie, Chinstrap, and Gentoo.

- **Body Mass (grams)**: The weight of the penguin, which serves as the response (dependent) variable.
- **Flipper Length (mm)**: The length of the penguin's flipper, used as a predictor.
- **Culmen Length (mm)**: The length of the penguin's bill (beak), measured along the top ridge, used as a predictor.
- **Culmen Depth (mm)**: The depth (height) of the penguin's bill, used as a predictor.
- **Species**: A categorical variable indicating the species of the penguin (Adelie, Chinstrap, or Gentoo), also used as a predictor.

The dataset initially contained a small number of missing values in some of the numerical variables. These observations were removed prior to analysis to ensure consistency and reliability of the regression models.

The variables selected for this study are biologically relevant, as body mass is expected to be influenced by structural features such as flipper size and bill dimensions, along with species-specific characteristics.

2.1 Summary Statistics

Variable	Mean	Minimum	Maximum	Std. Dev.
Body Mass (g)	4201.754	2700	6300	801.9545
Flipper Length (mm)	200.9152	172	231	14.06171
Culmen Length (mm)	43.92193	32.1	59.6	5.459584
Culmen Depth (mm)	17.15117	13.1	21.5	1.974793

Table 1: Overall Summary Statistics

Species	Variable	Mean	Minimum	Maximum	Std. Dev.
Adelie (151)	**Body Mass (g)**	3700.662	2850	4775	458.5661
	Flipper Length (mm)	189.9536	172	210	6.539457
	Culmen Length (mm)	38.79139	32.1	46	2.663405
	Culmen Depth (mm)	18.34636	15.5	21.5	1.21665
Gentoo (123)	**Body Mass (g)**	5076.016	3950	6300	504.1162
	Flipper Length (mm)	217.187	203	231	6.484976
	Culmen Length (mm)	47.50488	40.9	59.6	3.081857
	Culmen Depth (mm)	14.98211	13.1	17.3	0.9812198
Chinstrap (68)	**Body Mass (g)**	3733.088	2700	4800	384.3351
	Flipper Length (mm)	195.8235	178	212	7.131894
	Culmen Length (mm)	48.83382	40.9	58	3.339256
	Culmen Depth (mm)	18.42059	16.4	20.8	1.135395

Table 2: Summary Statistics of Species

3 Model Formulation

3.1 Theoretical Assessment

Biological traits (such as beak dimensions, flipper length, and body mass) scale proportionally relative to one another, a phenomenon known as allometry. This proportional, multiplicative growth is mathematically represented by a non-linear power law equation: $Y = \alpha X^{\beta}$.

Ordinary Least Squares (OLS) regression can only fit strictly linear relationships. By applying a natural logarithm to both sides of the allometric power law, we algebraically transform the biological curve into a perfectly straight line:

$$\log(Y) = \log(\alpha) + \beta \log(X).$$

This mathematical linearization allows the standard OLS model to accurately capture and estimate the proportional scaling rate (β) as a simple additive slope.

3.2 Statistical Assessment

First, we briefly introduce categorical variables before putting them to use. Unlike continuous variables (such as Flipper Length or Body Mass) which represent measurable, numerical quantities, categorical variables represent qualitative groups or labels (such as

Penguin Species). Ordinary Least Squares (OLS) regression relies on strictly numerical equations. Therefore, categorical variables must be mathematically transformed before they can be included in a linear model.

To incorporate qualitative data, statistical models utilize Dummy Variables (also known as One-Hot Encoding). This process translates categorical labels into a series of binary variables, assigning a 1 if the condition is met and a 0 if it is not.

In context of our current dataset, we are going to use the following categorical variables:

- `Is_Gen`: 1 if Gentoo, 0 otherwise
- `Is_Chin`: 1 if Chinstrap, 0 otherwise.

The baseline, Adelie, is mathematically implied when both `Is_Gentoo` and `Is_Chinstrap` are both 0. We do not account for this separately in the model as it's expected value is absorbed directly into the model's intercept, β_0.

Our objective is to predict the response variable, `Body_M`, using the predictors `Flip_L`, `Culm_L`, `Culm_D`, `Is_Gent`, and `Is_Chin`. We have $n = 342$ row entries in our dataset and $p = 5$ many parameters. Clearly $n > p + 1$. Furthermore, if we let X be the $n \times (p + 1)$ order matrix where the first column has all entries 1 and the other columns have the row-wise entries of the value of the predictors stated above, in order, then X has full column rank. Therefore, our standing assumptions for multiple linear regression have been verified.

We shall now verify the credibility of the normality assumption of the errors. First, we fit a model using the raw data from our dataset and then compare it with the model fitted by using the log-transformed data.

We know that $\hat{\beta}$ is given by $(X^\mathsf{T} X)^{-1} X^\mathsf{T} Y$. Plugging in the values for the raw data, we get that

$$\hat{\beta}_{\text{raw}} = (-4327.327, 20.241, 41.468, 140.328, 934.887, -513.247)^\mathsf{T}$$

for which the corresponding fitted model is

$$\begin{aligned}\widehat{\texttt{Body_M}} = -4327.327 + 20.241 \cdot \texttt{Flip_L} + 41.468 \cdot \texttt{Culm_L} + 140.328 \cdot \texttt{Culm_D} \\ + 934.887 \cdot \texttt{Is_Gen} - 513.247 \cdot \texttt{Is_Chin}.\end{aligned}$$

So, the residuals, i.e., $\widehat{\varepsilon_i}$, are given by $\texttt{Body_M}_i - \widehat{\texttt{Body_M}_i}$. Two plots of the residuals for the raw model have been provided below:

Figure 1: Plot of Residuals vs. Observation Index for Raw Model

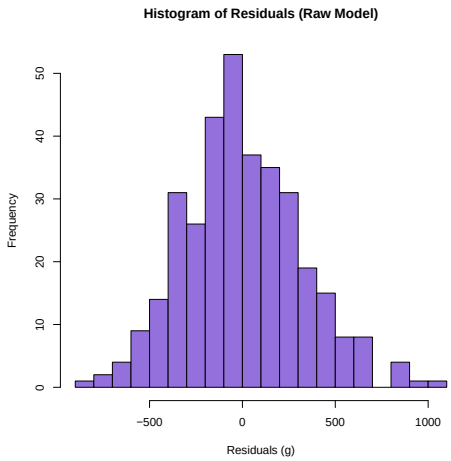

Figure 2: Histogram of Residuals for the Raw Model

As it can be noted in the histogram plot of the residuals, the distribution is skewed towards the right. Thus, this might pose problems for the normality assumption.

Now, let us work with the log-transformed model. Proceeding similarly before, we get

$$\hat{\beta}_{\log} = (0.07590, 0.90358, 0.43644, 0.61806, 0.23408, -0.11952)^{\mathsf{T}}$$

for which the corresponding fitted model is

$$\begin{aligned}\widehat{\log(\texttt{Body_M})} = 0.07590 &+ 0.90358 \cdot \log(\texttt{Flip_L}) + 0.43644 \cdot \log(\texttt{Culm_L}) \\ &+ 0.61806 \cdot \log(\texttt{Culm_D}) + 0.23408 \cdot \texttt{Is_Gen} - 0.11952 \cdot \texttt{Is_Chin}.\end{aligned}$$

In this case, the residuals, i.e., $\widehat{\varepsilon}_i$, are given by $\log(\texttt{Body_M}_i) - \log(\widehat{\texttt{Body_M}_i})$. Two plots of the residuals for the log model have been provided below:

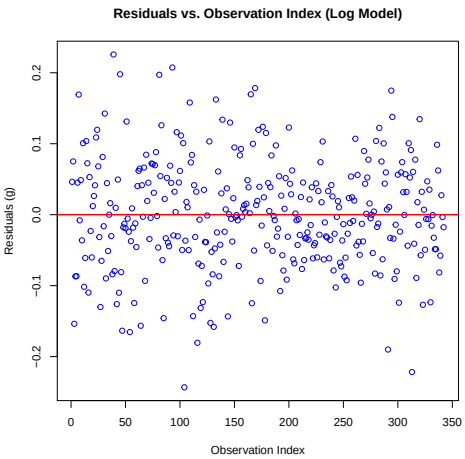

Figure 3: Plot of Residuals vs. Observation Index for Log Model

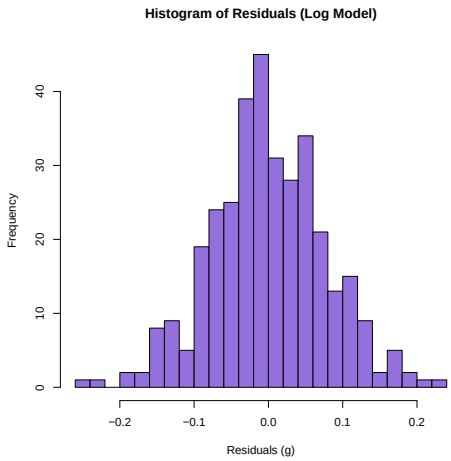

Figure 4: Histogram of Residuals for the Log Model

While the histogram of the raw model looks like a nice bell curve, it stretches out a little bit further in the right tail than the left tail. However, in the histogram of the log-transformed model, the curve looks almost perfectly and the left and right tails mirror each other around the center zero mark.